Rauf Shah 11/19/2025

Public Health Training/Teaching Project: Synchronous Distance Course Outline
Title: Cancer Surveillance: From Data to Public Health Action

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Course Information

Course: Public Health Teaching and Training (CPH 705)

Session Type: Synchronous (Zoom/Teams)

Duration: 60 minutes

Instructor: Rauf Shah, MPH-EPI

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1. Session Overview

This live session helps learners see how cancer registry data travels from the chart to public health action. We will quickly walk through key pieces-surveillance types, coding and staging, and data quality-and then apply them in a breakout case on a "Rural Spike" in lung cancer to decide if the pattern is real or a data artifact.

2. Learning Objectives

By the end of this 60-minute session, learners will be able to:

- Briefly define and tell apart passive, active, sentinel, and syndromic surveillance.
- Use basic coding and staging rules (ICD-O-3, TNM, reportability) to classify cancer cases.
- Identify which data quality dimension (completeness, accuracy, timeliness, consistency) is most at risk in a given scenario.
- Draw a simple workflow showing cancer data from case finding to public reporting.

3. Materials & Technology

- Slide deck: **Cancer Surveillance - Synchronous Session** (Slides 1-20).
- Zoom or Teams with polls and breakout rooms (3-4 learners per room).
- Student handout: **Skeleton notes**.
- Activity sheet: **"The Rural Spike" case study**.
- Web resources (shared in chat or on slides):
  - SEER Training - https://training.seer.cancer.gov/
  - NAACCR Standards - https://www.naaccr.org/data-standards-data-dictionary/

4. Session Outline (60 Minutes)

Time	Slides	Topic	Instructor Actions	Learner Engagement
1 min	1	Welcome & my Intro	Brief self-introduction and welcome.	Greet in chat; confirm audio/Zoom is working.
4 min	2-4	Session Roadmap & Baseline Poll	Outline goals, format, and activities; launch confidence poll on passive vs active surveillance.	Answer baseline confidence poll; respond to “Where do you work/learn?” in chat.
10 min	5-10	Burden & Approaches	Show key lung vs breast incidence/mortality slides; link to screening and early detection.	Chat: one observation from the graph; answer “Why is lung cancer mortality so high?”
10 min	11-13	Rules & Data Ecosystem	Brief overview of ICD-O-3 and TNM; walk through data flow diagram; highlight vital records.	Chat/voice: respond about where data enter and why death certificates matter.
5 min	14-15	Data Quality Dimensions	Introduce completeness, accuracy, timeliness, consistency; show NPCR targets.	Chat: answer “Which dimension suffers most with only passive reporting?”
15 min	16	Breakout - “The Rural Spike”	Present scenario and instructions; share activity link; open breakout rooms.	In groups: decide real vs artifact, pick dimension at risk, propose one data check; choose spokesperson.
10 min	16-18	Debrief, Security & Reporting	Bring class back; collect group diagnoses in chat; summarize; briefly cover security and small-cell suppression.	Post group diagnosis/check; respond to “Why is small-cell suppression important in rural areas?”
5 min	19-20	Wrap-Up & Q&A	Recap key points; share links/next steps; invite final questions.	Ask questions; share one key takeaway and note follow-up tasks.

5. Engagement Activities

Activity 1 - Confidence Poll (Slide 4)

- Question: *"How confident are you in defining passive vs active surveillance?"*
    - Very confident (I could teach it)
    - Somewhat confident (I know the basics)
    - Not confident (New or unclear to me)
- The poll results guide how much time we spend on the "Types of Surveillance" section.

Activity 2 - Breakout: "The Rural Spike" (Slide 16)

- **Scenario:** A rural county shows a 20% rise in late-stage lung cancer cases over 12 months, around the same time a local hospital switches to a new EHR.
- In breakout rooms, learners will:
    - **Analyze:** Could this be a real increase or a data artifact?
    - **Diagnose:** Which data quality dimension is most at risk?
    - **Solve:** Suggest one concrete data check (e.g., re-abstract a few cases, compare pre/post-EHR coding, or review staging rules) to help sort it out.

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